ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE D · MARKING GUIDELINE

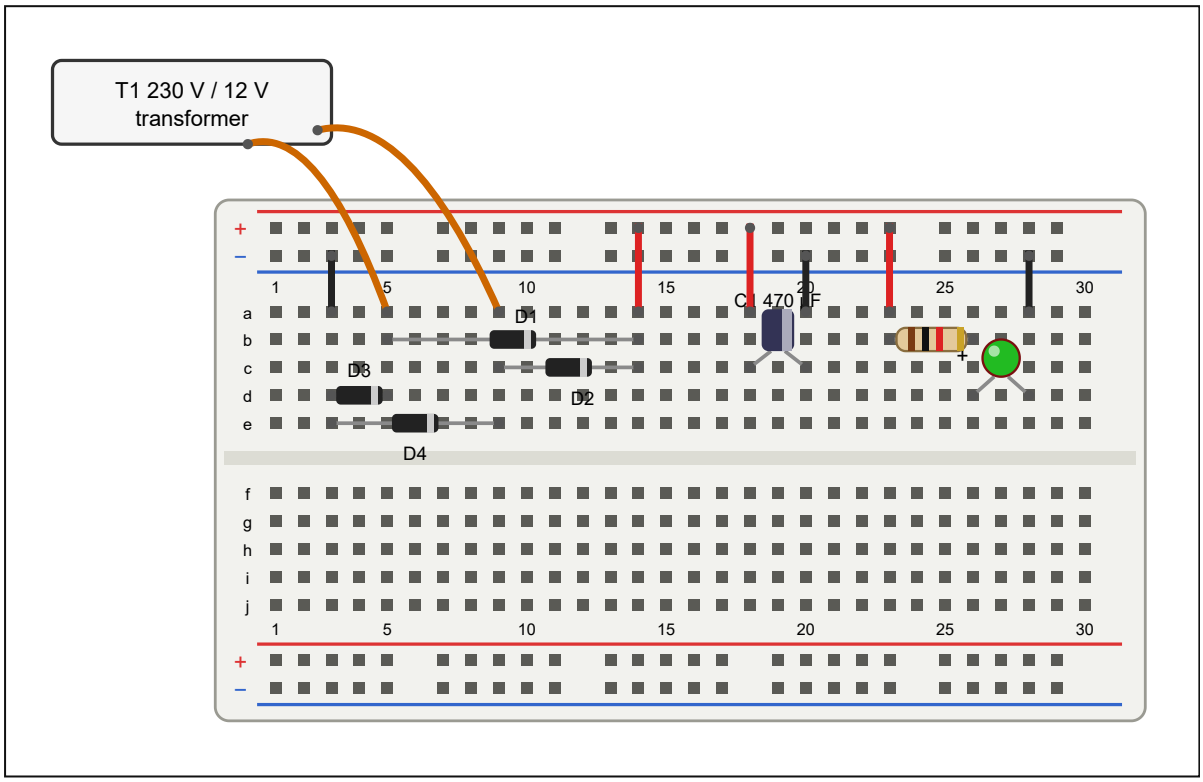
MARKS: 80

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics paper 2 · practice d.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Breadboard layouts may look different and still be correct: check that each connection in the circuit is made. Deduct 1 mark for each wrong connection and 2 marks for any short circuit (no less than 0 for a question).

QUESTION 1

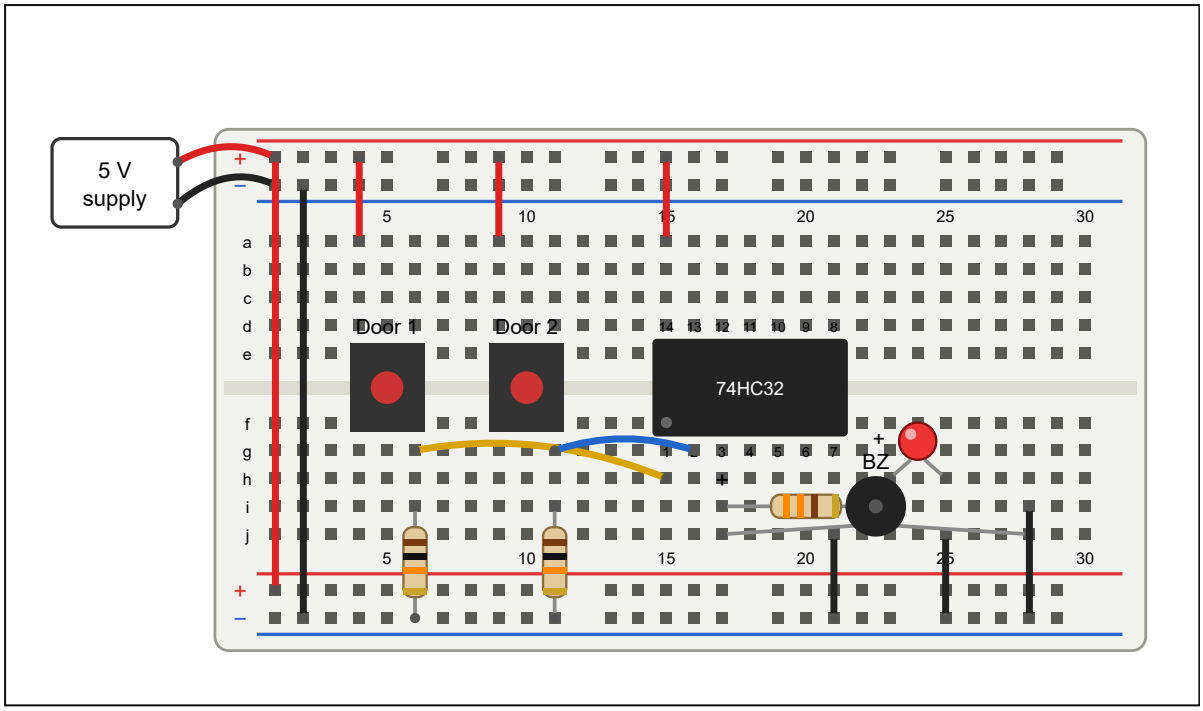


Example of a correct breadboard layout (other neat, correct layouts get full marks)

Placement ✓✓✓✓✓✓✓ (transformer, D1, D2, D3, D4, C1, R1, LED) = 8. Polarity ✓✓✓✓: all four diodes facing the correct way (cathodes of the two "top" diodes to +) ✓✓; C1 + to the + rail ✓; LED anode towards R1 ✓ = 4. Connections ✓✓✓✓✓✓✓✓: transformer secondary to the two AC points of the bridge ✓✓; bridge + output to the + rail ✓; bridge − output to the − rail ✓; C1 across the rails ✓; R1 from + to the LED ✓; LED cathode to − ✓; 230 V side clearly separate / labelled ✓ = 8. (20)

[20]

QUESTION 2

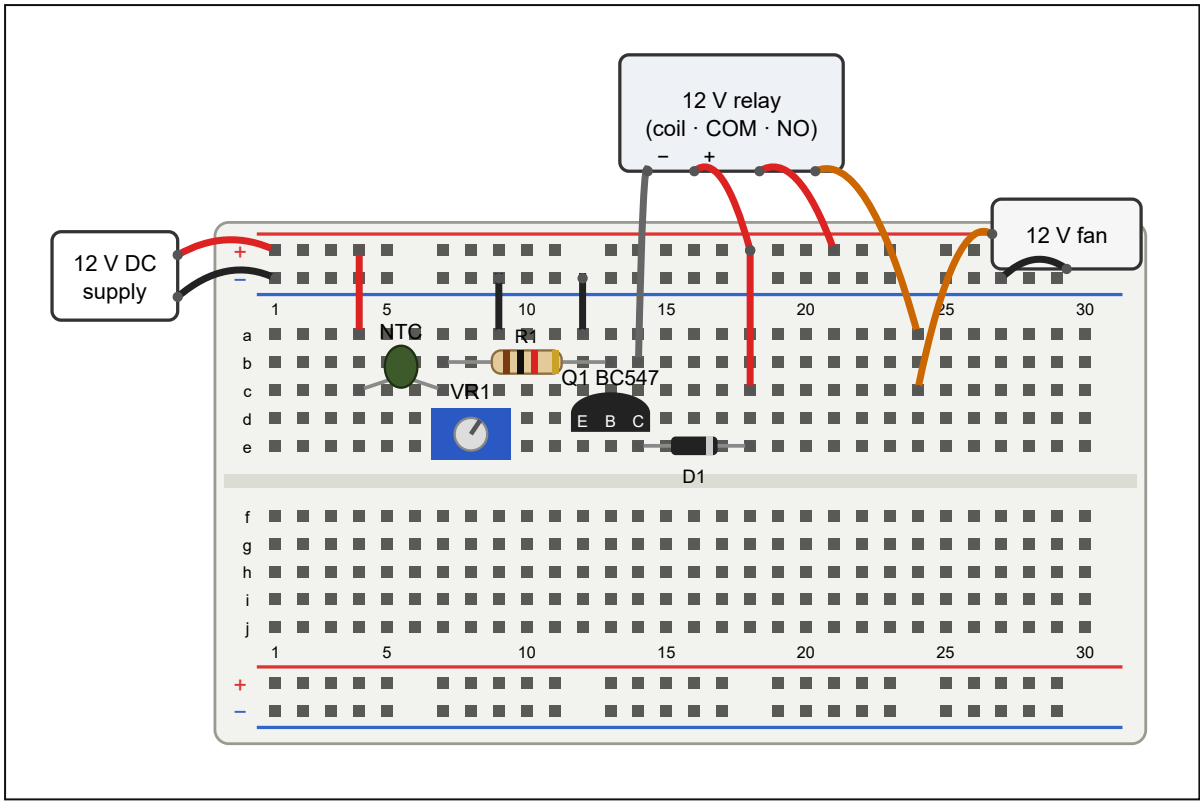


Example of a correct breadboard layout (other neat, correct layouts get full marks)

Placement: IC across the centre gap ✓; two reed switches / buttons ✓✓; R1 ✓; R2 ✓; R3 ✓; LED ✓; buzzer ✓ = 8. Polarity: IC notch / pin 1 correct ✓; LED ✓; buzzer + towards pin 3 ✓ = 3. Connections ✓✓✓✓✓✓✓✓✓: pin 14 to +5 V; pin 7 to 0 V; door 1 switch from +5 V to pin 1; door 2 switch from +5 V to pin 2; R1 from pin 1 to 0 V; R2 from pin 2 to 0 V; pin 3 to R3 and LED to 0 V; pin 3 to the buzzer and buzzer to 0 V; supply rails connected = 9. (20)

[20]

QUESTION 3



Example of a correct breadboard layout (other neat, correct layouts get full marks)

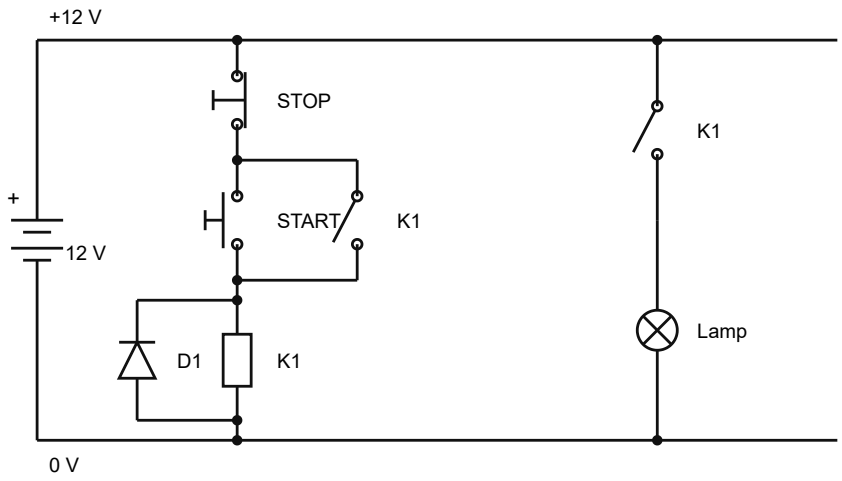
Correct circuit: thermistor from +12 V to the divider point; potentiometer from the divider point to 0 V; divider point through the 1 kΩ resistor to the base; emitter to 0 V; relay coil from +12 V to the collector; diode across the coil with its cathode to +12 V; relay COM to +12 V and NO contact to the fan +; fan − to 0 V.

Marks, 2 each: 12 V supply with polarity ✓✓; thermistor at the top of the divider ✓✓; potentiometer at the bottom ✓✓; 1 kΩ base resistor ✓✓; BC547 with correct pins, emitter to 0 V ✓✓; relay coil in the collector circuit ✓✓; diode across the coil the right way ✓✓; relay contacts switching the fan ✓✓; fan polarity ✓✓; neat layout and printout with name / examination number ✓✓ (20)

[20]

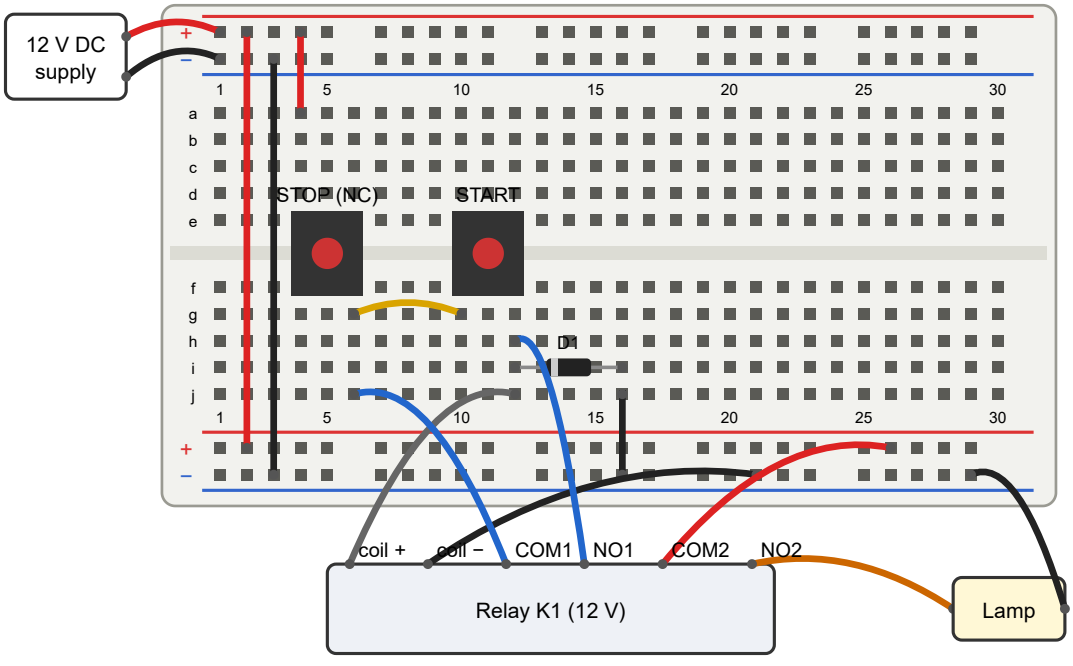
QUESTION 4

4.1



(+12 V and 0 V lines 1; STOP (NC) in series with START (NO) 1; relay contact K1 (NO) in parallel with START for latching 1; relay coil with the diode across it, cathode to + 1; second K1 contact switching the lamp across the supply 1)

(5)



Example of a correct breadboard layout for 4.2

4.2 Placement ✓✓✓✓✓✓ (supply, START, STOP, relay, diode, lamp) = 6. Polarity ✓✓: diode stripe towards +12 V; supply rails = 2. Connections ✓✓✓✓✓✓✓: +12 V to STOP; STOP to START; START to the relay coil; holding contact across START; coil to 0 V; second contact from +12 V to the lamp; lamp to 0 V = 7. (15)

[20]

TOTAL: 80

END OF MARKING GUIDELINE